Informatica Interview Prep

This is a copy of a conversation between ChatGPT & Anonymous.

> I am having interview on informatica power center. I will give questions.give me answer in simple and easy with example

Sure 👍
You can send the **Informatica PowerCenter interview questions**, and I will give:

- **Simple explanation**
- **Easy to understand**
- **Short definition**
- **Example (real-time if possible)**

I'll format answers like this so it's **easy to remember in interviews**:

1. Definition
2. Explanation
3. Example

Example:

Question: What is ETL?

Answer:

1. Definition:
ETL stands for **Extract, Transform, Load**. It is the process of moving data from source systems to a target system like a data warehouse.

2. Explanation:

- **Extract** – Get data from source (database, file, API).
- **Transform** – Clean, filter, or change the data.
- **Load** – Store the processed data into the target database.

3. Example:
A company stores customer data in **MySQL**.
They use **Informatica PowerCenter** to:

- **Extract** customer data from MySQL
- **Transform** by removing duplicate customers
- **Load** into a **Data Warehouse** for reporting.

Send your **questions one by one or all together**, and I'll prepare **interview-ready answers** for you. 🚀

> What is informatica power center

What is Informatica PowerCenter?

1. Definition:
Informatica PowerCenter is an **ETL tool** used to **extract data from different sources, transform the data, and load it into a target system** like a data warehouse or database.

2. Explanation:
It helps companies **integrate and process large amounts of data** from different systems such as:

- Databases (Oracle, MySQL, SQL Server)
- Flat files
- XML files
- Cloud sources

PowerCenter performs operations like:

- **Filtering data**
- **Removing duplicates**
- **Joining multiple tables**
- **Data cleansing**
- **Loading into target database**

It is mainly used in **Data Warehousing and Business Intelligence projects**.

3. Example:
Suppose a retail company has customer data in **three different databases**.

Using **PowerCenter**:

1. **Extract** customer data from Oracle and MySQL.
2. **Transform** the data by removing duplicate customers and formatting phone numbers.
3. **Load** the clean data into a **data warehouse** for reporting.

Short Interview Answer:

> Informatica PowerCenter is an ETL tool used to extract data from different sources, transform the data based on business rules, and load it into a target system like a data warehouse for reporting and analysis.

Informatica power center server services

Informatica PowerCenter Server Services

In **Informatica PowerCenter**, server services are the **background services that run and manage ETL operations** like workflows, sessions, and data processing.

1. Repository Service

Definition:
Repository Service manages the **PowerCenter repository database**.

Explanation:

- Stores all **metadata** such as mappings, workflows, sessions, and transformations.
- Allows developers to **save, retrieve, and manage objects**.

Example:
When you create a **mapping to load customer data**, it is stored in the **repository** and managed by the Repository Service.

---

2. Integration Service

Definition:
Integration Service is responsible for **executing workflows and sessions**.

Explanation:

- Reads the mapping and workflow from the repository.
- Extracts data from source.
- Applies transformations.
- Loads data into the target.

Example:
If you run a workflow to **load customer data from a flat file to Oracle**, the Integration Service performs the ETL process.

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3. Reporting Service (sometimes used in BI environments)

Definition:
Provides **reporting information about workflows, sessions, and jobs**.

Example:
A manager can check **which workflow succeeded or failed** using reports generated by this service.

---

4. PowerCenter Client Tools (used with services)

Although not server services, these tools interact with them:

- **Designer** – Create mappings and transformations
- **Workflow Manager** – Create workflows and sessions
- **Workflow Monitor** – Monitor workflow execution

Simple Interview Answer

> Informatica PowerCenter has two main server services: **Repository Service** and **Integration Service**. Repository Service manages the metadata stored in the repository, while Integration Service executes workflows and performs the ETL process like extracting, transforming, and loading data.

Explain domain and node

Domain in Informatica

In **Informatica PowerCenter**, a **Domain** is the **main administrative unit** that manages all Informatica services and nodes.

Explanation:

- It is the **top level container** in Informatica.
- It manages **security, services, and nodes**.
- A domain can contain **multiple nodes and multiple services**.

Example:
Suppose a company installs Informatica on two servers.
Both servers are connected under **one domain** called **Informatica_Domain**, which controls all services like **Repository Service** and **Integration Service**.

Simple interview answer:

> A Domain is the top-level administrative unit in Informatica that manages nodes, services, security, and configuration.

Node in Informatica

In **Informatica PowerCenter**, a **Node** is a **machine or server within the domain where Informatica services run**.

Explanation:

- A node represents a **physical or virtual machine**.
- Services like **Repository Service** or **Integration Service** run on nodes.
- A domain can have **one or multiple nodes**.

Example:

- **Node1** → Server where Repository Service runs
- **Node2** → Server where Integration Service runs

Both nodes are part of the **same domain**.

Simple interview answer:

A Node is a machine or server in an Informatica domain where services like Repository Service and Integration Service are installed and run.

Very Short Difference (Good for Interview)

Domain	Node
Top level administrative unit	Physical/virtual machine
Manages services and nodes	Runs Informatica services
Can contain multiple nodes	Belongs to one domain

If you want, I can also explain **Repository vs Domain vs Node** (this is a **very common Informatica interview question**).

> Explain client server services

In **Informatica PowerCenter**, the architecture works in a **Client–Server model**.

Client–Server Services in Informatica PowerCenter

1. Client

Definition:
Client is the **front-end tool used by developers to design ETL processes**.

Explanation:
Developers use client tools to:

- Create **mappings**
- Create **workflows**
- Monitor ETL jobs

Main Client Tools:

- **Designer** – Create mappings and transformations
- **Workflow Manager** – Create workflows and sessions
- **Workflow Monitor** – Monitor workflow execution
- **Repository Manager** – Manage repository objects

Example:
A developer uses **Designer** to create a mapping that loads customer data from a flat file to a database.

---

2. Server

Definition:
Server is the **backend system that executes the ETL processes** created by the client.

Explanation:
The server runs services that:

- Read mappings from the repository
- Extract data from source
- Transform data
- Load data into the target

Main Server Services:

- **Repository Service** – Manages metadata in the repository
- **Integration Service** – Executes workflows and performs ETL processing

Example:
After a developer creates a workflow in **Workflow Manager**, the **Integration Service** on the server executes it and loads data into the target database.

Simple Real-Time Example

A company wants to load **customer data from a CSV file to Oracle database**.

1. Developer creates **mapping in Designer** (Client).
2. Creates **workflow in Workflow Manager** (Client).
3. Workflow is stored in the **Repository**.
4. **Integration Service (Server)** runs the workflow and loads data into Oracle.

Short Interview Answer

> Informatica PowerCenter follows a client-server architecture. Client tools like Designer and Workflow Manager are used by developers to design mappings and workflows, while server services like Repository Service and Integration Service execute and manage the ETL processes.

Explain designer, workflow monitor,manager, repository manager

In **Informatica PowerCenter**, these are **client tools** used by developers to design, manage, and monitor ETL processes.

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1. Designer

Definition:
Designer is used to **create ETL mappings and transformations**.

Explanation:
In Designer, developers:

- Import **source tables**
- Create **target tables**
- Build **mappings**
- Apply **transformations** like Filter, Joiner, Aggregator, Expression.

Example:
A developer creates a mapping to:

- Read customer data from a **flat file**
- Remove inactive customers
- Load the data into an **Oracle table**.

---

2. Workflow Manager

Definition:
Workflow Manager is used to **create and manage workflows**.

Explanation:
It helps to:

- Create **sessions**
- Create **workflows**
- Schedule ETL jobs
- Define task order.

Example:
A workflow may contain:

- Session1 → Load customer data
- Session2 → Load order data

Workflow Manager controls the **execution order**.

---

3. Workflow Monitor

Definition:
Workflow Monitor is used to **monitor workflow and session execution**.

Explanation:
It shows:

- Workflow **status** (Running, Failed, Succeeded)
- Session logs
- Error messages
- Performance details

Example:
If a workflow fails due to **database connection error**, you can check the error in **Workflow Monitor**.

---

4. Repository Manager

Definition:
Repository Manager is used to **manage repository objects and folders**.

Explanation:
It is used to:

- Create **folders**
- Manage **users and permissions**
- Import or export objects
- Maintain repository.

Example:
A company may create folders like:

- **Sales_Project**
- **Customer_Project**

Each folder stores related mappings and workflows.

Short Interview Answer

> Informatica PowerCenter has several client tools. **Designer** is used to create mappings and transformations. **Workflow Manager** is used to create workflows and sessions. **Workflow Monitor** is used to monitor workflow execution and check errors. **Repository Manager** is used to manage repository objects, folders, and user permissions.

How many transformations are there in informatica power center. What are they

In **Informatica PowerCenter**, there are **around 16 main transformations** used to manipulate and process data during ETL.

Below are the **common transformations** with a simple explanation.

Transformations in Informatica PowerCenter

1. Expression Transformation

Used to **perform calculations or modify data**.

Example:
Concatenate first name and last name → Full Name.

---

2. Filter Transformation

Used to **filter rows based on a condition**.

Example:
Load only customers where **CUST_ACTIVE = 'Y'**.

---

3. Aggregator Transformation

Used to **perform aggregate functions** like SUM, COUNT, AVG.

Example:
Find **total sales for each country**.

---

4. Joiner Transformation

Used to **join data from two different sources**.

Example:
Join **Customer table** with **Orders table** using CUSTOMER_ID.

---

5. Lookup Transformation

Used to **look up data from another table**.

Example:
Find **customer city from customer table** using CUSTOMER_ID.

---

6. Router Transformation

Used to **split data into multiple groups** based on conditions.

Example:
Send **Active customers** to one target and **Inactive customers** to another.

---

7. Rank Transformation

Used to **select top or bottom records**.

Example:
Top **10 customers with highest sales**.

---

8. Sorter Transformation

Used to **sort data** in ascending or descending order.

Example:
Sort customers by **CUSTOMER_ID**.

---

9. Sequence Generator

Used to **generate unique numbers**.

Example:
Generate **Customer ID automatically**.

---

10. Update Strategy

Used to **insert, update, delete, or reject records** in the target.

Example:
Update customer address if it already exists.

---

11. Union Transformation

Used to **combine data from multiple pipelines**.

Example:
Combine customer data from **two different files**.

---

12. Normalizer

Used to **convert repeated columns into rows** (mainly for COBOL data).

---

13. Stored Procedure Transformation

Used to **call a stored procedure from the database**.

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14. Source Qualifier

Used to **read data from the source database**.

---

15. Transaction Control

Used to **control commit and rollback operations**.

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16. XML Transformation

Used to **process XML data**.

Short Interview Answer

> Informatica PowerCenter has around **16 transformations**. Some commonly used transformations are **Expression, Filter, Aggregator, Joiner, Lookup, Router, Rank, Sorter, Sequence Generator, Update Strategy, Union, Normalizer, Stored Procedure, Source Qualifier, Transaction Control, and XML Transformation**.

If you want, I can also tell you **the 10 most important transformations interviewers usually ask with examples** (very common question).

> Source qualifier

Source Qualifier Transformation

In **Informatica PowerCenter**, **Source Qualifier** is a **default transformation used to read data from the source database**.

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1. Definition

Source Qualifier is an **active transformation that retrieves rows from the source tables** and passes them to other transformations in the mapping.

---

2. Explanation

When you import a **source table** into a mapping, **Source Qualifier is automatically created**.

It is used to perform operations like:

- **Filtering rows**
- **Sorting data**
- **Joining multiple source tables**
- **Selecting specific columns**
- **Writing SQL queries**

---

3. Example

Suppose you have a **CUSTOMER table** in a database.

CUSTOMER_ID	NAME	COUNTRY
101	Ravi	India

CUSTOMER_ID	NAME	COUNTRY
102	John	USA
103	Priya	India

If you want to load **only customers from India**, you can write a condition in the **Source Qualifier SQL**:

```
SELECT * FROM CUSTOMER
WHERE COUNTRY = 'India'
```

Then only **Ravi and Priya** will be loaded.

---

4. Key Points (Interview)

- It is an **Active transformation**
- It is **automatically created** when source is added
- Used to **read data from source database**
- Allows **SQL override, filtering, sorting, and joining**

Short Interview Answer

> Source Qualifier is an active transformation in Informatica PowerCenter that reads data from the source database and passes it to other transformations. It is automatically created when a source is added to a mapping and can be used for filtering, sorting, joining, and SQL override.

Is source qualifier used for data type conversion based on according to information power center

Short Answer:
Yes, Source Qualifier can perform limited data type conversion in some cases.

In **Informatica PowerCenter**, the **Source Qualifier** mainly **reads data from the source**, but it can also **handle automatic datatype conversion when the source datatype and Informatica datatype are different but compatible**.

Explanation

When data is read from the source database:

- Informatica converts the **source datatype to an Informatica datatype internally**.
- If the **target datatype is compatible**, the conversion happens automatically.

This conversion happens **during the Source Qualifier stage** while reading the data.

However, **complex datatype conversions are usually done using an Expression Transformation**, not in Source Qualifier.

Example

Source table:

CUSTOMER_ID	ORDER_DATE
101	12-03-2025

- In source → **ORDER_DATE = VARCHAR**
- In target → **ORDER_DATE = DATE**

When the data passes through **Source Qualifier**, Informatica may convert it into the **Informatica Date format** if it is compatible.

If the format is different, then we use **Expression Transformation**, for example:

```
TO_DATE(ORDER_DATE,'DD-MM-YYYY')
```

Important Interview Point

Source Qualifier can perform:

- **Filtering**
- **Sorting**
- **Join**
- **SQL override**
- **Limited datatype conversion**

Simple Interview Answer

> Source Qualifier mainly reads data from the source database. It can perform limited datatype conversion when the source and target datatypes are compatible, but complex datatype conversions are usually done using Expression Transformation.

Give some examples that which datatype is converted to other data type in power center

In **Informatica PowerCenter**, some **datatype conversions happen automatically** when data moves from **source** → **Informatica** → **target**, if the datatypes are compatible.

Below are some **common examples**.

Examples of Datatype Conversion in Informatica PowerCenter

1. Integer → Decimal

Source: INTEGER
Target: DECIMAL

Example

Source Table

EMP_ID
100

Target Column datatype → **DECIMAL(10,2)**

Result in target:

EMP_ID
100.00

---

2. Integer → String

Source: INTEGER
Target: VARCHAR

Example

Source value

ORDER_ID
500

Target datatype → **VARCHAR**

Result stored as:

ORDER_ID
"500"

---

3. String → Date

Source: VARCHAR
Target: DATE

Example

Source value

ORDER_DATE
15-03-2026

Using expression:

```
TO_DATE(ORDER_DATE,'DD-MM-YYYY')
```

Result in target → **DATE format**

---

4. Decimal → Integer

Source: DECIMAL
Target: INTEGER

Example

Source value

PRICE
150.75

Target integer column will store:

PRICE
150

(decimal part removed)

---

5. String → Integer

Source: VARCHAR
Target: INTEGER

Example

Source value

AGE
"25"

Using expression:

```
TO_INTEGER(AGE)
```

Result:

AGE
25

Common Conversion Functions in Informatica

- **TO_CHAR()** → Convert to string
- **TO_DATE()** → Convert to date
- **TO_INTEGER()** → Convert to integer
- **TO_DECIMAL()** → Convert to decimal

Simple Interview Answer

> In Informatica PowerCenter, compatible datatypes can be converted automatically. Examples include Integer to Decimal, Integer to String, Decimal to Integer, and String to Date. For complex conversions, functions like TO_DATE, TO_CHAR, and TO_INTEGER are used in Expression Transformation.

Default converter in source qualifier

Default Conversion in Source Qualifier

In **Informatica PowerCenter**, the **Source Qualifier performs default datatype conversion automatically** when data is read from the source database.

Definition

Default conversion means **Informatica automatically converts the source datatype to the Informatica internal datatype** when the data enters the mapping through the **Source Qualifier**.

How it Works

When the Source Qualifier reads data:

1. It checks the **source database datatype**.
2. It converts it into **Informatica native datatype**.
3. The data then moves to the next transformation.

Example

Source Table (Oracle)

Column	Source Datatype
EMP_ID	NUMBER
EMP_NAME	VARCHAR2
JOIN_DATE	DATE

After passing through **Source Qualifier**, Informatica converts them to:

Column	Informatica Datatype
EMP_ID	Integer / Decimal
EMP_NAME	String
JOIN_DATE	Date/Time

This is called **default conversion**.

Important Interview Points

- Happens **automatically in Source Qualifier**.
- Converts **database datatype** → **Informatica datatype**.
- No manual expression is required for compatible datatypes.

Short Interview Answer

> In Informatica PowerCenter, Source Qualifier performs default datatype conversion automatically when it reads data from the source database. It converts the source database datatype into the Informatica internal datatype before passing the data to other transformations.

What are the ports in transformation

In **Informatica PowerCenter**, **ports** are the **input and output fields of a transformation** through which data flows.

Ports in Transformation

Definition

A **port** is a **column or field in a transformation that receives or sends data**.

When data passes through a transformation, it moves **through ports**.

Types of Ports in Informatica

1. Input Port

Definition:
A port that **receives data from the previous transformation or source**.

Example:
If a mapping reads **CUSTOMER_ID** from the source, it enters the transformation through an **Input Port**.

---

2. Output Port

Definition:
A port that **sends data to the next transformation or target**.

Example:
After processing, **CUSTOMER_NAME** is sent to the target table through an **Output Port**.

---

3. Input/Output Port

Definition:
A port that **both receives data and passes it to the next transformation**.

Example:
In an **Expression Transformation**, the column **SALARY** comes as input and passes unchanged to the next step.

---

4. Variable Port

Definition:
A port used to **store intermediate calculations inside a transformation**.

Example:

```
v_total = PRICE * QUANTITY
o_total = v_total + TAX
```

Here **v_total** is a **Variable Port**.

Example in Expression Transformation

Port Name	Type	Expression
PRICE	Input	-
QUANTITY	Input	-
v_total	Variable	PRICE * QUANTITY
o_total	Output	v_total

Simple Interview Answer

> In Informatica PowerCenter, ports are the fields inside a transformation through which data flows. There are four types of ports: Input Port, Output Port, Input/Output Port, and Variable Port. They are used to receive, process, and send data between transformations.

Give an example for variable port with code

Example of Variable Port in Informatica

In **Informatica PowerCenter**, a **Variable Port** is used to **store intermediate calculation results** inside a transformation (usually in an **Expression Transformation**).

Example: Calculate Total Salary

Suppose a source table has:

EMP_ID	BASIC_SALARY	BONUS
101	20000	3000

We want to calculate **Total Salary = BASIC_SALARY + BONUS + TAX**.

Expression Transformation Ports

Port Name	Port Type	Expression
BASIC_SALARY	Input	-
BONUS	Input	-
TAX	Input	-
v_temp_salary	Variable	BASIC_SALARY + BONUS
o_total_salary	Output	v_temp_salary + TAX

Code (Expression)

```
v_temp_salary = BASIC_SALARY + BONUS
o_total_salary = v_temp_salary + TAX
```

Result

If:

BASIC_SALARY = 20000
BONUS = 3000
TAX = 2000

Then:

v_temp_salary = 23000
o_total_salary = 25000

Key Interview Point

- Variable port **stores intermediate values**
- It **cannot pass data outside the transformation**

- It is used **only inside the transformation logic**

Short Interview Answer

> A Variable Port in Informatica is used to store intermediate calculation results inside a transformation. For example, in an Expression Transformation we can first calculate a temporary value in a variable port and then use it in the output port to produce the final result.

What is mapping

What is Mapping in Informatica?

In **Informatica PowerCenter**, **Mapping** is the **process of defining how data moves from source to target using transformations**.

Definition

A **mapping** is a **set of source tables, target tables, and transformations connected together to perform ETL operations**.

Explanation

In a mapping, we:

1. **Read data from source**
2. **Apply transformations** (filter, join, calculate, etc.)
3. **Load the processed data into the target**

Mapping defines the **data flow and transformation logic**.

Example

Source Table: **CUSTOMER**

CUSTOMER_ID	NAME	COUNTRY
101	Ravi	India
102	John	USA

Requirement: Load **only Indian customers** into the target table.

Mapping Flow:

Source → **Source Qualifier** → **Filter Transformation (COUNTRY='India')** → **Target**

Result in Target:

CUSTOMER_ID	NAME	COUNTRY
101	Ravi	India

Key Points

- Created in **Designer Tool**
- Defines **source** → **transformation** → **target**
- Contains **transformations, ports, and links**

Short Interview Answer

> Mapping in Informatica PowerCenter is a logical representation of the ETL process that defines how data is extracted from source, transformed using different transformations, and loaded into the target.

What is workflow

What is Workflow in Informatica?

In **Informatica PowerCenter**, a **Workflow** is a **set of instructions that tells Informatica how and when to run a mapping**.

Definition

A **workflow** is a **collection of tasks such as sessions, commands, or email tasks that are executed in a defined order to run the ETL process**.

Explanation

After creating a **mapping**, it cannot run directly.
We must create a **workflow** to execute that mapping.

A workflow controls:

- **Execution order of tasks**
- **Scheduling**
- **Error handling**
- **Monitoring**

Example

Suppose a company wants to load **customer data into a data warehouse**.

Steps in Workflow:

1. **Session Task** → Run Customer Mapping
2. **Session Task** → Run Order Mapping
3. **Email Task** → Send notification if job fails

Workflow Structure:

Start → Session(Customer Mapping) → Session(Order Mapping) → End

---

Key Points

- Created in **Workflow Manager**
- Executes **mappings through sessions**
- Controls **ETL job execution**

Short Interview Answer

> A Workflow in Informatica PowerCenter is a set of instructions that defines how and when ETL tasks like sessions are executed to run mappings and load data from source to target.

What is session

What is Session in Informatica?

In **Informatica PowerCenter**, a **Session** is a **task that runs a mapping** to move data from source to target.

Definition

A **Session** is a **workflow task that executes a mapping and performs the ETL process**.

Explanation

When a mapping is created, it cannot run directly.
A **Session is created inside a Workflow** to run that mapping.

The session controls:

- **Source and target connections**
- **Commit intervals**
- **Error handling**
- **Performance settings**

Example

Suppose we created a mapping called:

m_Load_Customer

To execute it:

Workflow → **Session Task (s_Load_Customer)** → Mapping runs

Flow:

Start → **Session (Run Mapping)** → Load Data → End

If the source table has:

CUSTOMER_ID	NAME
101	Ravi

The session runs the mapping and loads this data into the **target table**.

Key Points

- Runs a **mapping**
- Created inside **Workflow Manager**
- Controls **runtime properties**

Short Interview Answer

> A Session in Informatica PowerCenter is a workflow task that runs a mapping. It executes the ETL process by extracting data from the source, applying transformations, and loading it into the target.

What session do

What does a Session do in Informatica?

In **Informatica PowerCenter**, a **Session executes a mapping and performs the ETL process**.

What a Session Does

A **Session** performs the following tasks:

1. **Reads mapping information** from the repository.
2. **Connects to source and target databases**.
3. **Extracts data from the source**.
4. **Applies transformations defined in the mapping**.

5. **Loads the processed data into the target table**.
6. **Generates session logs and status** (success or failure).

Example

Suppose there is a mapping:

m_Load_Customer

Source Table:

CUSTOMER_ID	NAME
101	Ravi
102	John

When the **Session runs**:

1. It reads the mapping **m_Load_Customer**.
2. Extracts data from the **source table**.
3. Applies transformations if any.
4. Loads the data into the **target table**.

Target Table Result:

CUSTOMER_ID	NAME
101	Ravi
102	John

Simple Interview Answer

> A Session in Informatica PowerCenter executes a mapping. It reads data from the source, applies transformations defined in the mapping, and loads the processed data into the target database while generating logs for monitoring.

What is workflow monitor containe

What does Workflow Monitor contain?

In **Informatica PowerCenter**, **Workflow Monitor** is used to **monitor and track the execution of workflows and sessions**.

Workflow Monitor contains the following components:

1. Workflow Runs

Shows the **status of workflows**.

Possible statuses:

- Running
- Succeeded
- Failed
- Aborted

Example:
Workflow **wf_Load_Customer** → Status: **Succeeded**

---

2. Session Runs

Displays **session execution details** inside the workflow.

It shows:

- Start time
- End time
- Rows processed
- Status

---

3. Task Details

Shows details of tasks such as:

- Session task
- Command task
- Email task
- Event task

---

4. Logs

Contains **session logs and workflow logs** used for **debugging errors**.

Example:
If a session fails due to **database connection error**, you can check the **session log** in Workflow Monitor.

---

5. Performance Details

Shows information like:

- Number of rows read from source
- Number of rows written to target

- Transformation statistics

Simple Interview Answer

> Workflow Monitor in Informatica PowerCenter is used to monitor workflow execution. It contains workflow runs, session runs, task details, logs, and performance information to check the status and troubleshoot errors.

Session log and workflow log

Session Log and Workflow Log in Informatica

In **Informatica PowerCenter**, logs are used to **track execution details and errors** during ETL processing.

---

1. Session Log

Definition

A **Session Log** contains **detailed information about the execution of a session (mapping run)**.

What it contains

- Source and target connection details
- Number of rows **read, written, rejected**
- Transformation details
- Errors and warnings
- Start time and end time

Example

Session **s_Load_Customer**

Session Log may show:

Detail	Value
Source Rows Read	1000
Target Rows Written	980
Rejected Rows	20

If a **datatype mismatch error occurs**, it will appear in the **session log**.

---

2. Workflow Log

Definition

A **Workflow Log** contains **information about the execution of the entire workflow**.

What it contains

- Workflow start and end time
- Execution order of tasks
- Status of each session
- Workflow errors

Example

Workflow: **wf_Load_Data**

Workflow Log may show:

Start → Session1 (Success) → Session2 (Failed) → Workflow Failed

---

Difference Between Session Log and Workflow Log

Session Log	Workflow Log
Shows session execution details	Shows workflow execution details
Contains mapping and transformation info	Contains task execution order
Used to debug data processing errors	Used to track workflow status

---

Short Interview Answer

> Session Log contains detailed information about the execution of a session such as rows processed, transformations, and errors. Workflow Log contains information about the overall workflow execution including task order, session status, and workflow start and end time.

Different between active and passive transformation

Difference Between Active and Passive Transformation

In **Informatica PowerCenter**, transformations are classified as **Active** and **Passive** based on how they affect the number of rows passing through them.

---

1. Active Transformation

Definition

An **Active Transformation** is a transformation that **can change the number of rows passing through it**.

What it can do

- **Filter rows**
- **Add new rows**
- **Join rows**
- **Aggregate rows**

Examples

- Filter Transformation
- Aggregator Transformation
- Joiner Transformation
- Router Transformation
- Rank Transformation
- Union Transformation

Example

Source Table:

ID	NAME
1	Ravi
2	John

Filter condition: **NAME = 'Ravi'**

Result:

ID	NAME
1	Ravi

Number of rows **changed**, so it is **Active**.

---

2. Passive Transformation

Definition

A **Passive Transformation** is a transformation that **does not change the number of rows passing through it**.

What it does

It only **modifies the data but keeps the same number of rows**.

Examples

- Expression Transformation
- Lookup Transformation (sometimes passive)
- Sequence Generator

Example

Source Table:

ID	SALARY
1	10000
2	20000

Expression: **SALARY + 1000**

Result:

ID	SALARY
1	11000
2	21000

Number of rows **remains the same**, so it is **Passive**.

Key Difference

Active Transformation	Passive Transformation
Changes number of rows	Does not change number of rows
Can filter, join, or aggregate data	Only modifies data
Example: Filter, Aggregator	Example: Expression

Short Interview Answer

> Active transformation can change the number of rows passing through it, while passive transformation does not change the number of rows and only modifies the data. Examples of active transformations are Filter and Aggregator, while Expression is a passive transformation.

Router transformation

Router Transformation

In **Informatica PowerCenter**, **Router Transformation** is used to **split data into multiple groups based on different conditions**.

Definition

Router Transformation is an **active transformation that routes data into multiple output groups according to specified conditions**.

Explanation

Router checks **each row against multiple conditions** and sends the row to the **appropriate output group**.

It has:

- **Input group**
- **Multiple output groups**
- **Default group** (for rows that do not match any condition)

Example

Source Table:

EMP_ID	DEPARTMENT
101	HR
102	IT
103	SALES

Router Conditions:

- **Group1 (HR_Group)** → DEPARTMENT = 'HR'
- **Group2 (IT_Group)** → DEPARTMENT = 'IT'
- **Default Group** → Others

Output:

HR Target

EMP_ID	DEPARTMENT
101	HR

IT Target

EMP_ID	DEPARTMENT
102	IT

Default Target

EMP_ID	DEPARTMENT
103	SALES

Key Points

- It is an **Active Transformation**
- Used to **route data into multiple targets**
- Supports **multiple conditions**
- Has a **default group**

Short Interview Answer

> Router Transformation is an active transformation in Informatica PowerCenter that splits data into multiple output groups based on different conditions. Rows that do not match any condition are sent to the default group.

What is the max no.of group we can create in router transformation

In **Informatica PowerCenter**, a **Router Transformation** can have **up to 100 output groups**.

Explanation

- Router has **one Input group**.
- It can have **multiple Output groups** where each group has a condition.
- It also has a **Default group** for rows that do not satisfy any condition.

So:

- **Maximum output groups = 100**
- **Default group = 1**

Example

If you want to separate employees by department:

- Group1 → HR
- Group2 → IT

- Group3 → SALES
- Group4 → FINANCE
- Default group → Others

Router checks the condition and **routes each row to the correct group**.

Short Interview Answer

> In Informatica PowerCenter Router Transformation, we can create a maximum of **100 output groups** along with **one default group**.

Filter transformation

Filter Transformation

In **Informatica PowerCenter**, **Filter Transformation** is used to **filter rows based on a condition**.

Definition

Filter Transformation is an **active transformation that removes rows that do not satisfy a specified condition**.

Explanation

- It checks each row against the **filter condition**.
- **Rows that satisfy the condition pass to the next transformation or target.**
- **Rows that do not satisfy the condition are discarded.**

Because some rows are removed, it is called an **Active Transformation**.

Example

Source Table:

CUSTOMER_ID	COUNTRY
101	India
102	USA
103	India

Filter Condition:

```
COUNTRY = 'India'
```

Result (Target Table):

CUSTOMER_ID	COUNTRY
101	India
103	India

Row with **USA** is filtered out.

Key Points

- It is an **Active Transformation**
- Used to **filter rows based on a condition**
- Rows not matching the condition are **dropped**

Short Interview Answer

> Filter Transformation is an active transformation in Informatica PowerCenter that filters records based on a specified condition and passes only the matching rows to the next transformation or target.

Aggregator transformation

Aggregator Transformation

In **Informatica PowerCenter**, **Aggregator Transformation** is used to **perform aggregate calculations on data**.

Definition

Aggregator Transformation is an **active transformation that performs aggregate functions such as SUM, COUNT, AVG, MAX, and MIN**.

Explanation

It groups data based on **Group By columns** and then performs calculations on those groups.

Common aggregate functions:

- **SUM()**
- **COUNT()**
- **AVG()**
- **MAX()**
- **MIN()**

Because it can **reduce the number of rows**, it is an **Active Transformation**.

Example

Source Table:

COUNTRY	SALES
India	100
USA	200
India	150
USA	100

Group By: **COUNTRY**

Expression:

```
TOTAL_SALES = SUM(SALES)
```

Result:

COUNTRY	TOTAL_SALES
India	250
USA	300

The rows are **grouped and aggregated**, so the number of rows **changes**.

Key Points

- It is an **Active Transformation**
- Used for **aggregate calculations**
- Requires **Group By** column for grouping data

Short Interview Answer

> Aggregator Transformation is an active transformation in Informatica PowerCenter used to perform aggregate calculations like SUM, COUNT, AVG, MAX, and MIN on grouped data.

Joiner transformation

Joiner Transformation

In **Informatica PowerCenter**, **Joiner Transformation** is used to **join data from two different sources or pipelines**.

Definition

Joiner Transformation is an **active transformation that joins data from two sources based on a common column (join condition)**.

Explanation

Joiner has two pipelines:

1. **Master source**
2. **Detail source**

Data is joined using a **join condition**, similar to SQL joins.

Common join types:

- **Inner Join**
- **Left Outer Join**
- **Right Outer Join**
- **Full Outer Join**

Example

Source 1: **CUSTOMER Table**

CUSTOMER_ID	NAME
101	Ravi
102	John

Source 2: **ORDER Table**

CUSTOMER_ID	ORDER_AMOUNT
101	500
102	700

Join Condition:

```
CUSTOMER.CUSTOMER_ID = ORDER.CUSTOMER_ID
```

Result:

CUSTOMER_ID	NAME	ORDER_AMOUNT
101	Ravi	500
102	John	700

Key Points

- It is an **Active Transformation**
- Used to **join data from two sources**
- Uses **Master and Detail pipelines**
- Supports **different join types**

Short Interview Answer

> Joiner Transformation is an active transformation in Informatica PowerCenter used to join data from two different sources based on a common column using join conditions such as inner join or outer join.

Explain the two pipeline

Two Pipelines in Joiner Transformation

In **Informatica PowerCenter**, **Joiner Transformation has two pipelines** through which data flows before it is joined.

These two pipelines are:

1. **Master Pipeline**
2. **Detail Pipeline**

---

1. Master Pipeline

Definition

The **Master Pipeline** is the **primary input of the Joiner transformation**.

Explanation

- Informatica **reads master data first**.
- It is usually the **smaller dataset** for better performance.

Example

Master Table: **CUSTOMER**

CUSTOMER_ID	NAME
101	Ravi
102	John

---

2. Detail Pipeline

Definition

The **Detail Pipeline** is the **secondary input that is joined with the master pipeline**.

Explanation

- Informatica compares the **detail records with master records** using the join condition.

Example

Detail Table: **ORDER**

CUSTOMER_ID	ORDER_AMOUNT
101	500
102	700

Join Condition

```
CUSTOMER.CUSTOMER_ID = ORDER.CUSTOMER_ID
```

Result After Join

CUSTOMER_ID	NAME	ORDER_AMOUNT
101	Ravi	500
102	John	700

Key Interview Points

- Joiner transformation **requires two pipelines**
- One is **Master**, the other is **Detail**
- Usually **smaller table is chosen as Master** for performance

---

Short Interview Answer

> In Joiner Transformation, there are two pipelines: Master pipeline and Detail pipeline. The master pipeline contains the primary dataset and the detail pipeline contains the secondary dataset. Informatica joins both pipelines using a join condition.

Explain the joiner transformation types

Joiner Transformation Types

In **Informatica PowerCenter**, **Joiner Transformation** supports **four types of joins** to combine data from two pipelines (Master and Detail).

---

1. Normal Join (Inner Join)

Explanation

Only the **matching rows from both Master and Detail pipelines** are returned.

Example

Master (Customer)

ID	NAME
1	Ravi
2	John

Detail (Order)

ID	AMOUNT
1	500
3	700

Result:

ID	NAME	AMOUNT
1	Ravi	500

Only **matching ID = 1** is returned.

---

2. Master Outer Join

Explanation

Returns **all rows from Master table** and matching rows from Detail table.

If there is no match, **Detail columns become NULL**.

Example Result

ID	NAME	AMOUNT
1	Ravi	500
2	John	NULL

3. Detail Outer Join

Explanation

Returns **all rows from Detail table** and matching rows from Master table.

If no match, **Master columns become NULL**.

Example Result

ID	NAME	AMOUNT
1	Ravi	500
3	NULL	700

4. Full Outer Join

Explanation

Returns **all rows from both Master and Detail pipelines**.

If there is no match, missing values are **NULL**.

Example Result

ID	NAME	AMOUNT
1	Ravi	500
2	John	NULL
3	NULL	700

Simple Summary

Join Type	Result
Normal (Inner) Join	Only matching rows
Master Outer Join	All master rows
Detail Outer Join	All detail rows
Full Outer Join	All rows from both tables

Short Interview Answer

> Joiner Transformation supports four join types: Normal Join (Inner Join), Master Outer Join, Detail Outer Join, and Full Outer Join. These joins determine how rows from master and detail pipelines are combined based on matching conditions.

Rank transformation

Rank Transformation

In **Informatica PowerCenter**, **Rank Transformation** is used to **select top or bottom records based on a specific column value**.

Definition

Rank Transformation is an **active transformation that selects the top or bottom N records from a dataset based on ranking criteria**.

Explanation

- It compares values in a column.
- Assigns **rank numbers** to rows.
- Returns only **Top N or Bottom N rows**.

It is **active** because it **changes the number of rows**.

Example

Source Table:

EMP_ID	SALARY
101	20000
102	30000
103	25000
104	40000

Requirement: **Top 2 highest salaries**

Ranking Column → **SALARY**
Rank Number → **2**

Result:

EMP_ID	SALARY
104	40000

EMP_ID	SALARY
102	30000

Key Points

- It is an **Active Transformation**
- Used to **select Top or Bottom records**
- Requires **Rank Port (column used for ranking)**
- Supports **Group By option**

Short Interview Answer

> Rank Transformation in Informatica PowerCenter is an active transformation used to select the top or bottom N records based on a specified column value such as highest salary or lowest sales.

Sequence transformation

Sequence Generator Transformation

In **Informatica PowerCenter**, **Sequence Generator Transformation** is used to **generate unique sequential numbers**.

Definition

Sequence Generator is a **passive transformation that generates unique numeric values for each row**.

Explanation

It automatically generates numbers like **1, 2, 3, 4...** for each record.
It is commonly used to **create primary keys or unique IDs**.

Main properties:

- **Start Value** – starting number
- **Increment By** – how much the number increases
- **End Value** – maximum value

Example

Source Table:

NAME
Ravi
John
Priya

Sequence Generator generates **Customer IDs**.

Start Value = **1**
Increment By = **1**

Result:

CUSTOMER_ID	NAME
1	Ravi
2	John
3	Priya

Important Ports

Sequence Generator has two output ports:

- **CURRVAL** → Current sequence value
- **NEXTVAL** → Next generated value

Example:

```
NEXTVAL = 1, 2, 3, 4...
```

Key Points

- Generates **unique sequential numbers**
- Used for **primary key generation**
- Has **CURRVAL and NEXTVAL ports**

Short Interview Answer

> Sequence Generator Transformation in Informatica PowerCenter is used to generate unique sequential numbers for each row, commonly used to create primary keys or unique IDs in the target table.

Difference between current and next value

Difference Between CURRVAL and NEXTVAL

In **Informatica PowerCenter**, the **Sequence Generator Transformation** has two important output ports:

- **CURRVAL**
- **NEXTVAL**

---

1. NEXTVAL (Next Value)

Definition

NEXTVAL generates the **next sequence number for every new row**.

Example

Start Value = **1**
Increment By = **1**

Output using **NEXTVAL**:

Row	NEXTVAL
1	1
2	2
3	3
4	4

Each row gets a **new unique value**.

---

2. CURRVAL (Current Value)

Definition

CURRVAL shows the **current value generated by NEXTVAL for that row**.

It returns the **same value generated in that cycle**.

Example

Row	NEXTVAL	CURRVAL
1	1	1
2	2	2
3	3	3

CURRVAL simply **holds the current generated value**.

Key Difference

CURRVAL	NEXTVAL
Shows current sequence value	Generates next sequence value
Used to reuse the same value in mapping	Used to create new sequence numbers

---

Simple Interview Answer

> In Sequence Generator, NEXTVAL generates the next sequential number for each row, while CURRVAL returns the current sequence value that was generated by NEXTVAL in the same row.

Why we map next value in mapping

Why do we map NEXTVAL in a Mapping?

In **Informatica PowerCenter**, we usually map **NEXTVAL** from the **Sequence Generator Transformation** because it **generates a new unique number for every row**.

Explanation

When data is loaded into a target table, we often need a **unique ID or primary key**.

NEXTVAL generates a **new sequence number for each record**, so every row gets a **unique value**.

That is why **NEXTVAL is mapped to the target column**.

Example

Source Table:

NAME
Ravi
John
Priya

Sequence Generator settings:

- Start Value = **1**
- Increment By = **1**

Mapping:

```
Sequence Generator (NEXTVAL) → CUSTOMER_ID
NAME → NAME
```

Target Table Result:

CUSTOMER_ID	NAME
1	Ravi
2	John
3	Priya

Here **NEXTVAL generates unique IDs for each row**.

Why we don't usually map CURRVAL

- **CURRVAL only holds the current value**
- It **does not generate a new sequence number**
- It is mainly used **inside the mapping if the same value needs to be reused**

Short Interview Answer

> We map NEXTVAL in a mapping because it generates a new unique sequence number for every row. It is commonly used to create primary keys or unique IDs when loading data into the target table.

Example for +2inc give for curr and next val

Example for Increment by +2 (CURRVAL and NEXTVAL)

In **Informatica PowerCenter**, when we configure a **Sequence Generator Transformation**, we can set the **Increment By value**.

Let's see an example with **Increment By = 2**.

Sequence Generator Settings

- **Start Value = 1**
- **Increment By = 2**

Generated Values

Row	NEXTVAL	CURRVAL
1	1	1
2	3	3
3	5	5
4	7	7

Explanation:

- First row → NEXTVAL = **1**
- Second row → NEXTVAL = **3** (1 + 2)
- Third row → NEXTVAL = **5** (3 + 2)

CURRVAL always shows the **current generated value of NEXTVAL** for that row.

Example Mapping

Source Table:

NAME
Ravi
John
Priya

Mapping:

```
Sequence Generator (NEXTVAL) → CUSTOMER_ID
NAME → NAME
```

Target Result:

CUSTOMER_ID	NAME
1	Ravi
3	John
5	Priya

Short Interview Answer

> If the increment value is set to 2 in Sequence Generator, NEXTVAL generates values like 1, 3, 5, 7. CURRVAL shows the current generated value for that row, which will be the same as NEXTVAL in that cycle.

Why we usually do not use curr val in mapping

Why we usually do not use CURRVAL in Mapping

In **Informatica PowerCenter**, **CURRVAL is rarely used in mappings** because it **does not generate new sequence numbers**.

Explanation

The **Sequence Generator** has two ports:

- **NEXTVAL** → Generates a **new unique number for every row**
- **CURRVAL** → Shows the **current value generated by NEXTVAL**

When loading data into a target table, we usually need **unique IDs for each record**.
So we map **NEXTVAL**, because it generates **new numbers for each row**.

Problem if we use CURRVAL

If **CURRVAL is used before NEXTVAL is generated**, it may give **incorrect or repeated values**.

Example:

Sequence Settings
Start Value = **1**
Increment = **1**

Row	NEXTVAL	CURRVAL
1	1	1
2	2	2
3	3	3

But **CURRVAL only reflects the value generated by NEXTVAL**, it does **not create a new value**.

When CURRVAL is used

CURRVAL is used **inside the mapping when we need to reuse the same generated value** in multiple places.

Example:

- Generate **Order_ID using NEXTVAL**
- Use **CURRVAL** to insert the same Order_ID into another related table.

Short Interview Answer

> We usually do not use CURRVAL in mappings because it does not generate new sequence numbers. NEXTVAL is used instead because it generates a unique value for each row. CURRVAL is mainly used when the same generated value needs to be reused within the mapping.

Lookup transformation

Lookup Transformation

In **Informatica PowerCenter**, **Lookup Transformation** is used to **get related data from another table or file based on a matching condition**.

Definition

Lookup Transformation is used to **look up data in a lookup table and return the corresponding value**.

Explanation

It compares a **column from the source with a column in the lookup table**.
If a match is found, it **returns the required column value**.

It is mainly used to:

- Get additional information
- Validate data
- Check if a record exists

Example

Source Table: **EMPLOYEE**

EMP_ID	DEPT_ID
101	10
102	20

Lookup Table: **DEPARTMENT**

DEPT_ID	DEPT_NAME
10	HR
20	IT

Lookup Condition:

```
EMPLOYEE.DEPT_ID = DEPARTMENT.DEPT_ID
```

Result:

EMP_ID	DEPT_ID	DEPT_NAME
101	10	HR
102	20	IT

The **Lookup Transformation fetches DEPT_NAME** from the department table.

Types of Lookup

1. **Connected Lookup**
2. **Unconnected Lookup**

Key Points

- Used to **retrieve data from another table**
- Works using a **lookup condition**
- Can be **active or passive depending on configuration**

Short Interview Answer

> Lookup Transformation in Informatica PowerCenter is used to retrieve related data from another table or file based on a matching condition between the source and the lookup table.

Explain the types

Types of Lookup Transformation

In **Informatica PowerCenter**, there are **two types of Lookup Transformations**:

1. **Connected Lookup**
2. **Unconnected Lookup**

---

1. Connected Lookup

Definition

A **Connected Lookup** is a lookup transformation that **is directly connected to the mapping pipeline**.

Explanation

- It receives **input directly from the source pipeline**.
- It returns **multiple columns** to the pipeline.
- It participates directly in the **data flow**.

Example

Source Table: **EMPLOYEE**

EMP_ID	DEPT_ID
101	10

Lookup Table: **DEPARTMENT**

DEPT_ID	DEPT_NAME
10	HR

Result after lookup:

EMP_ID	DEPT_ID	DEPT_NAME
101	10	HR

The **DEPT_NAME** column is returned through the pipeline.

---

2. Unconnected Lookup

Definition

An **Unconnected Lookup** is a lookup transformation that **is not directly connected to the pipeline**.

Explanation

- It is **called using a function inside an expression**.
- It returns **only one column value**.

Example

Expression:

```
:lkp.DEPT_NAME(DEPT_ID)
```

Here:

- **DEPT_ID** is passed as input.
- Lookup returns **DEPT_NAME**.

Result:

EMP_ID	DEPT_ID	DEPT_NAME
101	10	HR

Key Differences

Connected Lookup	Unconnected Lookup
Connected to pipeline	Not connected to pipeline
Returns multiple columns	Returns one column

Connected Lookup	Unconnected Lookup
Direct data flow	Called using lookup function

Short Interview Answer

> Lookup Transformation has two types: Connected Lookup and Unconnected Lookup. Connected Lookup is directly connected to the mapping pipeline and can return multiple columns, while Unconnected Lookup is not connected to the pipeline and is called using a lookup function that returns a single value.

What is the difference between joiner,union and lookup transformation. When to use these

In **Informatica PowerCenter**, **Joiner, Union, and Lookup transformations** are used for different purposes when combining or retrieving data.

Difference Between Joiner, Union, and Lookup

Transformation	Purpose	When to Use
Joiner	Combines columns from two sources based on a common key	When you need data from two tables with a matching column
Union	Combines rows from multiple sources with the same structure	When you want to merge data from multiple pipelines
Lookup	Retrieves related data from another table	When you need to get additional information from another table

---

1. Joiner Transformation

Use

Used to **join two sources using a common column**.

Example

Customer Table

ID	NAME
1	Ravi

Order Table

ID	AMOUNT
1	500

Join Condition:

```
CUSTOMER.ID = ORDER.ID
```

Result:

ID	NAME	AMOUNT
1	Ravi	500

Use Joiner when you want to combine columns from two different tables.

---

2. Union Transformation

Use

Used to **combine rows from multiple sources into one output**.

Example

Source 1

NAME
Ravi
John

Source 2

NAME
Priya
David

Result after Union

NAME
Ravi
John
Priya
David

Use Union when sources have the same structure and you want to merge rows.

---

3. Lookup Transformation

Use

Used to **retrieve additional information from another table**.

Example

Employee Table

EMP_ID	DEPT_ID
101	10

Department Table

DEPT_ID	DEPT_NAME
10	HR

Lookup returns:

EMP_ID	DEPT_ID	DEPT_NAME
101	10	HR

Use Lookup when you need to fetch a value from another table.

Simple Way to Remember (Interview Trick)

- **Joiner → Combine columns from two sources**
- **Union → Combine rows from multiple sources**
- **Lookup → Retrieve value from another table**

Short Interview Answer

> Joiner transformation is used to combine data from two sources based on a join condition. Union transformation is used to merge rows from multiple sources with the same structure. Lookup transformation is used to retrieve additional data from another table based on a lookup condition.

Union transformation

Union Transformation

In **Informatica PowerCenter**, **Union Transformation** is used to **combine data from multiple pipelines into a single pipeline**.

Definition

Union Transformation is an **active transformation that merges rows from two or more sources with the same structure into one output**.

Explanation

- It combines **rows from multiple input pipelines**.
- The **number of columns and datatype must be the same** in all pipelines.
- It works similar to **UNION in SQL**, but it **does not remove duplicate rows automatically**.

Example

Source 1: **EMPLOYEE_INDIA**

EMP_ID	NAME
101	Ravi
102	Priya

Source 2: **EMPLOYEE_USA**

EMP_ID	NAME
201	John
202	David

After **Union Transformation**

Result:

EMP_ID	NAME
101	Ravi
102	Priya
201	John
202	David

Rows from **both sources are merged into one output**.

Key Points

- It is an **Active Transformation**

- Combines **rows from multiple sources**
- Input pipelines must have **same number of columns and datatype**

Short Interview Answer

> Union Transformation in Informatica PowerCenter is an active transformation used to combine rows from multiple input pipelines into a single output pipeline, provided the input structures are the same.

Why it is active

Why Union Transformation is Active

In **Informatica PowerCenter**, **Union Transformation is an Active Transformation** because it **changes the number of rows that pass through the pipeline**.

Explanation

An **active transformation** is one that **can change the number of rows** in the data flow.

The **Union Transformation** takes **rows from multiple input pipelines** and **combines them into one output pipeline**.

Because it **increases the number of rows**, it is considered **active**.

Example

Source 1:

ID	NAME
1	Ravi
2	John

Source 2:

ID	NAME
3	Priya
4	David

After **Union Transformation**:

ID	NAME
1	Ravi
2	John
3	Priya

ID	NAME
4	David

Here:

- Input pipelines had **2 rows each**
- Output pipeline has **4 rows**

So the **number of rows changed**, which makes it **Active**.

Short Interview Answer

> Union Transformation is active because it combines rows from multiple input pipelines and changes the total number of rows in the output.

What are the conditions that we have to check for active transformation

Conditions to Identify an Active Transformation

In **Informatica PowerCenter**, a transformation is called **Active** if it **changes the data flow in the pipeline**.

Main Conditions

A transformation is **Active** if it performs any of these actions:

1. Changes the Number of Rows

It may **increase, decrease, or filter rows**.

Example:
Filter Transformation

Source:

ID	NAME
1	Ravi
2	John

Condition:

```
ID > 1
```

Result:

ID	NAME
2	John

Rows changed from **2** → **1**.

---

2. Changes the Transaction Boundary

It can **control commit or rollback operations**.

Example: **Transaction Control Transformation**

---

3. Changes the Row Type

It may **mark rows as insert, update, delete, or reject**.

Example: **Update Strategy Transformation**

---

4. Changes the Row Order

Some transformations **sort or rearrange rows**.

Example: **Sorter Transformation**

Examples of Active Transformations

- Filter
- Aggregator
- Joiner
- Router
- Rank
- Union
- Sorter
- Update Strategy

Short Interview Answer

> A transformation is considered active in Informatica PowerCenter if it changes the number of rows, modifies transaction boundaries, changes row types, or alters the order of rows in the pipeline.

Check in check out object

Check-In and Check-Out Object

In **Informatica PowerCenter**, **Check-In and Check-Out** is used to **control access to repository objects like mappings, workflows, and transformations** when multiple developers work on the same project.

Check-Out

Definition

Check-Out means **locking an object so that you can edit it**.

Explanation

- When you check out an object, **only you can modify it**.
- Other developers **cannot edit the same object** until you check it back in.

Example

If you want to modify a **mapping**, you must first **check out the mapping**.

Check-In

Definition

Check-In means **saving the changes and releasing the lock on the object**.

Explanation

- After editing the object, you **check it in**.
- Now other developers can **access and modify the object**.

Example Scenario

Developer A checks out **Mapping1**.

Result:

- Developer A → Can edit Mapping1
- Developer B → Cannot edit Mapping1

After Developer A **checks in Mapping1**:

- Developer B can now **edit it**.

Key Points

- Used in **Repository Manager**

- Prevents **multiple users editing the same object**
- Maintains **version control**

Short Interview Answer

> Check-Out is used to lock an object in the repository so a developer can modify it, and Check-In is used to save the changes and release the lock so other developers can access the object.

Give a sinario based example

Scenario-Based Example of Check-In and Check-Out in Informatica

Imagine a **team of 3 developers** working on a **Customer Data Warehouse project** in **Informatica PowerCenter**.

Scenario

Objects in repository:

- Mapping: `m_Load_Customers`
- Workflow: `wf_Load_Customers`

Team Members:

- Developer A
- Developer B
- Developer C

Step 1: Check-Out

- Developer A wants to **update the mapping** `m_Load_Customers` to add a new **calculated column**.
- Developer A **checks out** the mapping.

Effect:

- Developer A → Can edit the mapping freely.
- Developer B & C → Cannot edit `m_Load_Customers`. They will see it as **locked**.

Step 2: Making Changes

- Developer A adds a new **expression port** in an Expression Transformation to calculate `Total_Sales`.
- Developer A tests the mapping in **Workflow Manager**.

Step 3: Check-In

- Once changes are complete, Developer A **checks in the mapping**.

Effect:

- The object is **unlocked**.
- Developer B can now check out the mapping to add **new transformation logic**, like handling `Discount`.
- Version history is maintained in the repository.

Key Benefits in Scenario

1. Prevents **conflict** when multiple developers work on the same object.
2. Maintains **version control** for ETL objects.
3. Ensures **team collaboration** is smooth.

Short Interview Answer

> For example, Developer A checks out the mapping `m_Load_Customers` to make changes. While checked out, others cannot edit it. After finishing changes, Developer A checks in the mapping so others can access it. This prevents conflicts and maintains version control.

If you want, I can also **draw a simple diagram showing Check-Out and Check-In flow** for this scenario—it helps in interviews. Do you want me to do that?

> Normalisation transformation

Normalizer Transformation

In **Informatica PowerCenter**, **Normalizer Transformation** is used to **normalize data, usually when working with COBOL sources or repeating groups**, and to generate **multiple rows from a single source row**.

Definition

Normalizer Transformation is a **passive transformation** that:

- Converts **repeating columns into multiple rows**
- Generates **sequence numbers** for each row

Explanation

- Often used when data comes from **COBOL sources** or **flat files** with repeating fields.
- Converts **wide data (many columns)** into **tall data (multiple rows)**.

- Can also generate **primary key sequences** using NEXTVAL.

Example

Source Table (COBOL-style):

CUSTOMER_ID	ORDER1	ORDER2	ORDER3
101	500	600	700

After **Normalizer Transformation**:

CUSTOMER_ID	ORDER_NUMBER	ORDER_AMOUNT
101	1	500
101	2	600
101	3	700

- One row with **3 orders** is normalized into **3 rows**.

Ports in Normalizer

1. **Input Ports** → Accept source data
2. **Output Ports** → Normalized data
3. **Sequence Port** → Can generate **sequence numbers**

Key Points

- Passive Transformation (does not filter rows unless configured)
- Converts **repeating groups into multiple rows**
- Can generate **sequence numbers**

Short Interview Answer

> Normalizer Transformation in Informatica PowerCenter is used to convert repeating columns in a single row into multiple rows and optionally generate sequence numbers. It is commonly used for COBOL sources or files with repeating groups.

If you want, I can also **give a simple scenario-based example with multiple customers and orders**, which is often asked in interviews. Do you want me to do that?

> Why it is passive

Why Normalizer Transformation is Passive

In **Informatica PowerCenter**, the **Normalizer Transformation is considered passive** because it **does not change the number of rows flowing through it in the sense of filtering or dropping rows—it only restructures them**.

Explanation

Passive Transformation Criteria:
A transformation is **passive** if:

1. It **does not filter rows**
2. It **does not change transaction boundaries**
3. It **does not mark rows as insert/update/delete**

Normalizer Transformation:

- Takes repeating fields from a single row and **splits them into multiple rows**, but **every value from the input is preserved**.
- It **does not filter or drop any row**.
- It just **reformats the data**, which is why it is **passive**.

Example

Source Table:

CUSTOMER_ID	ORDER1	ORDER2
101	500	600

After Normalizer:

CUSTOMER_ID	ORDER_NUMBER	ORDER_AMOUNT
101	1	500
101	2	600

- Input data is **fully preserved**, nothing is removed.
- It **restructures rows**, but does not "actively" filter or delete them.

Short Interview Answer

> Normalizer Transformation is passive because it does not filter or remove rows. It only restructures repeating fields into multiple rows while preserving all input data.

Gcid and gkvalue in this

GCID and GKVAL in Normalizer Transformation

In **Informatica PowerCenter**, the **Normalizer Transformation** automatically generates **special system ports**: **GCID** and **GKVAL**. These are used to **identify groups of repeating fields** and their sequence numbers.

---

1. GCID (Group Context ID)

Definition:

- GCID is a **unique ID for each repeating group instance**.
- It **identifies which input row a normalized output row belongs to**.

Example:

Source Table:

CUSTOMER_ID	ORDER1	ORDER2
101	500	600

Normalizer Output:

CUSTOMER_ID	ORDER_NUMBER	ORDER_AMOUNT	GCID
101	1	500	1
101	2	600	1

- Both rows belong to the same input row, so **GCID = 1**.

---

2. GKVAL (Group Key Value)

Definition:

- GKVAL is the **sequence number or occurrence of a repeating field within the group**.
- Shows the **position of the value in the original repeating group**.

Example:

CUSTOMER_ID	ORDER_NUMBER	ORDER_AMOUNT	GCID	GKVAL
101	1	500	1	1
101	2	600	1	2

- **GKVAL = 1** for ORDER1, **GKVAL = 2** for ORDER2

Simple Summary

Port	Meaning
GCID	Identifies the original input row (group)
GKVAL	Identifies the position of a repeating field within the group

---

Short Interview Answer

> In Normalizer Transformation, GCID is a unique ID for each input row (group) to which the normalized rows belong, and GKVAL is the sequence number of the repeating field within that group, showing its position.

If you want, I can also **draw a small table showing source data → normalized data → GCID & GKVAL clearly**, which is very helpful for interviews. Do you want me to do that?

> why we use normalization transformation

Why We Use Normalizer Transformation

In **Informatica PowerCenter**, the **Normalizer Transformation** is used for **handling repeating data and creating multiple rows from a single row**.

Main Reasons to Use Normalizer

1. **Handling Repeating Groups**
   - Some sources (like COBOL files, flat files, or legacy systems) store data in **repeating columns**.
   - Normalizer **converts these repeating columns into separate rows**.
   - Example: Orders stored as `ORDER1, ORDER2, ORDER3` in one row → converted to multiple rows.
2. **Creating Multiple Rows**
   - Normalizer **splits one input row into multiple output rows**.
   - Useful when each repeated value should be treated as a separate record.
3. **Generating Sequence Numbers**
   - Can generate **unique IDs or sequence numbers** for each normalized row using **NEXTVAL**.
   - Helps in creating **primary keys** for target tables.
4. **Preparing Data for Aggregation or Lookup**
   - Normalized data is easier to **join, aggregate, or look up** in later transformations.

- Makes ETL pipelines cleaner and more structured.

Example

Source Table:

CUSTOMER_ID	ORDER1	ORDER2
101	500	600

After Normalizer:

CUSTOMER_ID	ORDER_NUMBER	ORDER_AMOUNT
101	1	500
101	2	600

- One row with **two orders** becomes **two rows**, which is easier to process in ETL.

Short Interview Answer

> We use Normalizer Transformation to convert repeating columns in a single row into multiple rows, generate sequence numbers, and prepare data for easier processing in ETL pipelines. It is especially useful for COBOL or flat file sources with repeating groups.

If you want, I can also **give a small real-life scenario with customers and multiple orders** which is often asked in interviews—it makes the explanation very clear. Do you want me to do that?

sorted transformation

Sorter Transformation

In **Informatica PowerCenter**, the **Sorter Transformation** is used to **sort data in ascending or descending order** based on one or more columns.

Definition

Sorter Transformation is an **active transformation** that **rearranges rows in a pipeline based on specified key columns**.

Explanation

- Can sort **one or more columns**
- Can sort **ascending or descending**

- Can perform **distinct sort** (remove duplicates)
- Can **group by** for aggregations (optional)

Because the **order of rows changes**, it is an **Active Transformation**.

Example

Source Table:

EMP_ID	SALARY
101	20000
102	30000
103	25000

Sort by **SALARY Descending**

Result:

EMP_ID	SALARY
102	30000
103	25000
101	20000

Key Points

- **Active Transformation** (row order changes)
- Can sort **ascending/descending**
- Can remove **duplicates** using distinct option
- Used before **aggregations, rank, or target load**

Short Interview Answer

> Sorter Transformation in Informatica PowerCenter is an active transformation used to sort data in ascending or descending order based on one or more columns. It can also remove duplicates using the distinct option.

If you want, I can also **explain why it's active vs passive with a quick example**, which is often asked in interviews. Do you want me to do that?

update statergy transformation

Update Strategy Transformation

In **Informatica PowerCenter**, the **Update Strategy Transformation** is used to **control how data is written to the target table**.

It tells Informatica whether the row should be:

- **Inserted**
- **Updated**
- **Deleted**
- **Rejected**

Why We Use Update Strategy

Sometimes when loading data to a target table, we need to decide what action to perform on each row.

For example:

- New record → **Insert**
- Existing record changed → **Update**
- Record removed → **Delete**

Update Strategy helps control these actions.

Example

Source Table

ID	NAME	STATUS
101	Ravi	NEW
102	John	UPDATE
103	David	DELETE

Expression in Update Strategy:

```
IIF(STATUS='NEW', DD_INSERT,
IIF(STATUS='UPDATE', DD_UPDATE,
IIF(STATUS='DELETE', DD_DELETE, DD_REJECT)))
```

Result:

ID	NAME	Action
101	Ravi	Insert
102	John	Update
103	David	Delete

Update Strategy Constants

Constant	Meaning
DD_INSERT	Insert row
DD_UPDATE	Update row
DD_DELETE	Delete row
DD_REJECT	Reject row

Why It Is Active

Update Strategy is **Active** because it **can reject rows or change row types**, which affects the pipeline flow.

Short Interview Answer

> Update Strategy Transformation is used to control how rows are written to the target table by specifying actions like insert, update, delete, or reject using predefined constants such as DD_INSERT and DD_UPDATE.

what does delete and reject does

What Do DELETE and REJECT Do in Update Strategy

In **Informatica PowerCenter**, the **Update Strategy Transformation** can mark rows for different actions such as **DD_DELETE** and **DD_REJECT**.

---

1. DD_DELETE

Meaning

DD_DELETE tells Informatica to **delete the existing record from the target table**.

Example

Target Table before loading:

ID	NAME
101	Ravi
102	John

Source Data:

ID	ACTION
102	DELETE

Expression:

```
IIF(ACTION='DELETE', DD_DELETE)
```

Result after loading:

ID	NAME
101	Ravi

Row **102 is removed from the target table**.

---

2. DD_REJECT

Meaning

DD_REJECT means the **row will not be loaded into the target table**.

The record is **discarded by the Integration Service**.

Example

Source Data:

ID	AGE
101	25
102	-5

Condition:

```
IIF(AGE < 0, DD_REJECT, DD_INSERT)
```

Result:

ID	AGE	Action
101	25	Insert
102	-5	Rejected

Row **102 is rejected and not loaded**.

Key Difference

DD_DELETE	DD_REJECT
Deletes record from target table	Discards row before loading

DD_DELETE	DD_REJECT
Works on existing target data	Row is not loaded at all

Short Interview Answer

> DD_DELETE removes an existing record from the target table, while DD_REJECT discards the row so it is not loaded into the target.

Import & Export

Import and Export in Informatica

In **Informatica PowerCenter**, **Import and Export** are used to **move repository objects between different repositories or environments**.

Export

Definition

Export means **saving Informatica objects from the repository into an XML file**.

Why we Export

- To **move objects to another repository**
- To **take backup of mappings, workflows, etc.**
- To **share objects with other developers**

Example

Suppose you created:

- Mapping → `m_Load_Customer`
- Workflow → `wf_Load_Customer`

You **export these objects**, and Informatica creates an **XML file**.

This file can be used later to import into another repository.

Import

Definition

Import means **loading objects from an XML file into the repository**.

Why we Import

- To **bring objects from another environment**
- To **restore backup**
- To **reuse mappings or workflows**

Example

Developer exports a mapping from **Development Repository**.
Another developer **imports the XML file** into **Test Repository**.

Simple Workflow Example

Developer A (Development Environment)

- Creates mapping
- **Exports → XML file**

Developer B (Testing Environment)

- **Imports XML file**
- Mapping is now available in their repository.

Key Points

Export	Import
Moves objects from repository to XML file	Moves objects from XML file to repository
Used for backup or migration	Used to reuse objects

Short Interview Answer

> Export is used to save Informatica repository objects like mappings and workflows into an XML file, while Import is used to load those objects from the XML file into another repository.

Mapplets , Worklets

Mapplet and Worklet

In **Informatica PowerCenter**, **Mapplets** and **Worklets** are reusable objects used to simplify ETL development.

---

1. Mapplet

Definition

A **Mapplet** is a **reusable set of transformations** used inside mappings.

Explanation

- It contains **multiple transformations grouped together**.
- It can be **reused in different mappings**.
- Helps reduce **duplicate transformation logic**.

Example

Suppose many mappings need to:

- Clean customer name
- Convert name to uppercase
- Remove special characters

Instead of creating these transformations in every mapping, we create a **Mapplet** called:

```
mp_Customer_Cleaning
```

Inside Mapplet:

- Expression Transformation → Remove special characters
- Expression Transformation → Convert to uppercase

Now this Mapplet can be used in **many mappings**.

---

2. Worklet

Definition

A **Worklet** is a **reusable set of workflow tasks** used inside workflows.

Explanation

- It contains **multiple tasks grouped together**.
- Used in **Workflow Manager**.
- Helps reuse **workflow logic**.

Example

Suppose many workflows need these tasks:

1. Start Task

2. Session Task
3. Email Notification

Instead of recreating them each time, we create a **Worklet** called:

`wk_Load_Process`

This Worklet can be used in **multiple workflows**.

Key Difference

Mapplet	Worklet
Used in **Mapping**	Used in **Workflow**
Contains **Transformations**	Contains **Tasks**
Created in **Designer**	Created in **Workflow Manager**

Short Interview Answer

> A Mapplet is a reusable set of transformations used inside mappings, while a Worklet is a reusable set of workflow tasks used inside workflows. Both help reduce redundancy and improve reusability.

Reusable Transformation

Reusable Transformation

In **Informatica PowerCenter**, a **Reusable Transformation** is a transformation that can be **used in multiple mappings**.

Definition

A **Reusable Transformation** is created **once in the repository** and can be **shared across different mappings**.

Why We Use It

- Avoid creating the **same transformation again and again**
- Maintain **common logic in one place**
- Easy to **update logic for all mappings**

Example

Suppose many mappings need to **convert customer name to uppercase**.

Instead of creating an **Expression Transformation** in every mapping:

We create one **Reusable Expression Transformation**

Name:
`exp_uppercase_name`

Logic:

```
UPPER(CUST_NAME)
```

Now this reusable transformation can be used in:

- Mapping 1 → `m_Load_Customer`
- Mapping 2 → `m_Load_Order`
- Mapping 3 → `m_Load_Employee`

Advantage

If we change the logic in the reusable transformation, **all mappings using it will automatically get updated**.

Key Points

- Created once in repository
- Can be used in **multiple mappings**
- Helps **reduce duplicate work**

Short Interview Answer

> A Reusable Transformation in Informatica PowerCenter is a transformation created once in the repository that can be used in multiple mappings, helping reduce redundancy and maintain common logic easily.

Parameter, session variable, mapping variable, workflow variable

Parameters and Variables

In **Informatica PowerCenter**, **parameters and variables** are used to **make mappings and workflows flexible and dynamic**.

---

1. Parameter

Definition

A **Parameter** is a **constant value that does not change during workflow execution**.

Example

Parameter:

```
$$SOURCE_FILE_PATH = /data/customer.txt
```

This value **remains the same during the whole workflow run**.

---

2. Session Variable

Definition

A **Session Variable** is used **inside a session to control session properties dynamically**.

Example

```
$$Commit_Interval = 10000
```

This variable controls **how many rows are committed during session execution**.

---

3. Mapping Variable

Definition

A **Mapping Variable** is a variable used **inside a mapping whose value can change during execution**.

Example

Suppose we want to store the **maximum salary processed**.

Expression:

```
SETVARIABLE($$MAX_SALARY, SALARY)
```

The value **updates during mapping execution**.

---

4. Workflow Variable

Definition

A **Workflow Variable** is used **inside a workflow to store values that can change during workflow execution**.

Example

```
$$Total_Rows_Processed
```

This variable can store the **number of rows processed by sessions**.

Key Differences

Type	Used In	Value Change
Parameter	Mapping / Session / Workflow	Constant during run
Session Variable	Session	Can change per session
Mapping Variable	Mapping	Changes during mapping execution
Workflow Variable	Workflow	Changes during workflow execution

Short Interview Answer

> A Parameter is a constant value during execution, while variables can change during runtime. Session variables are used in sessions, mapping variables are used inside mappings, and workflow variables are used within workflows to store dynamic values.

give a example for all of these Parameter, session variable, mapping variable, workflow variable

Examples for Parameter, Session Variable, Mapping Variable, Workflow Variable

In **Informatica PowerCenter**, parameters and variables help make ETL processes flexible and dynamic.

---

1. Parameter Example

Scenario

The **source file path** may change in different environments.

Parameter:

```
$$SOURCE_FILE_PATH=/data/customer_file.txt
```

Mapping uses this parameter to read the file.

Explanation:
The value **remains constant during workflow execution**.

---

2. Session Variable Example

Scenario

We want to control how often data is committed to the database.

Session Variable:

```
$$COMMIT_INTERVAL=10000
```

Meaning:

- After **every 10,000 rows**, Informatica commits the data.

---

3. Mapping Variable Example

Scenario

We want to **store the maximum salary processed in the mapping**.

Variable:

```
$$MAX_SALARY
```

Expression logic:

```
SETVARIABLE($$MAX_SALARY, SALARY)
```

Example data:

EMP_ID	SALARY
101	20000
102	30000

Result:

-

---

4. Workflow Variable Example

Scenario

We want to **count total rows processed in the workflow**.

Workflow Variable:

```
$$TOTAL_ROWS
```

Session updates the variable with the number of rows processed.

Example:

Session	Rows Processed
Session1	5000

Workflow Variable value:

```
$$TOTAL_ROWS = 5000
```

Simple Difference Example

Type	Example Use
Parameter	Source file path
Session Variable	Commit interval

Type	Example Use
Mapping Variable	Max salary processed
Workflow Variable	Total rows processed

Short Interview Answer

> A parameter stores a constant value like source file path, a session variable controls session properties like commit interval, a mapping variable stores values that change during mapping execution such as maximum salary, and a workflow variable stores dynamic values used during workflow execution like total rows processed.

what is informatica powercenter and when it was released,created

What is Informatica PowerCenter

Informatica PowerCenter is an **ETL (Extract, Transform, Load) tool** used to **extract data from different sources, transform the data, and load it into a target system such as a data warehouse or database**.

It helps organizations **integrate and process large amounts of data from multiple systems**.

Simple Example

Suppose a company has data in:

- **Oracle database** (customer data)
- **Flat file** (sales data)

Using **Informatica PowerCenter**:

1. **Extract** data from Oracle and flat file
2. **Transform** data (clean, filter, convert formats)
3. **Load** data into a **Data Warehouse**

Who Created It

It was created by **Informatica**.

When It Was Released

- **Company Founded:** 1993
- **PowerCenter Released:** Around **1997**

Short Interview Answer

Informatica PowerCenter is an ETL tool used to extract data from different sources, transform the data according to business rules, and load it into a target system like a data warehouse. It was developed by Informatica and released around 1997.